UNIVERSITY OF ILLINOIS AT URBANA-CHAMPAIGN
Department of Electrical and Computer Engineering

ECE 310 Digital Signal Processing – Fall 2026

# Homework 1

**Profs. Kamalabadi, Snyder** **Due: Friday, Sep 4, 11:59pm on Gradescope**

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1. Plot each of the following discrete-time signals for $-5 \leq n \leq 5$. Recall that $\delta[n]$ is the unit impulse or Kronecker delta and $u[n]$ is the unit step signal.

   (a) $\delta[n+4] - \delta[n+1] + 3\delta[n] + 2\delta[n-2]$

   (b) $\left(\frac{2}{3}\right)^n (u[n+2] - u[n-3])$

   (c) $nu[n]u[n-4]$

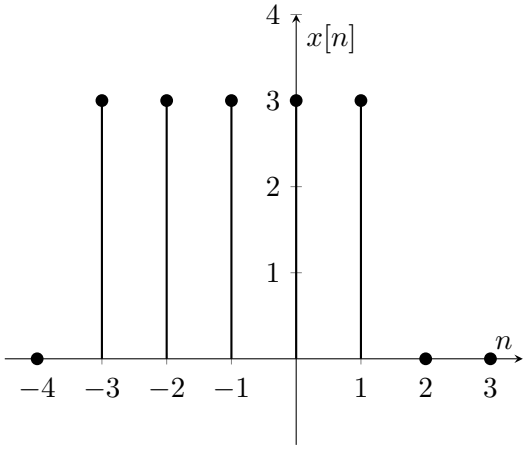
2. Let $x[n]$ be a length–8 discrete-time signal as given below.

$$x[n] = \{2, 4, \underset{\uparrow}{1}, 0, -2, 5, 7, 3\}$$

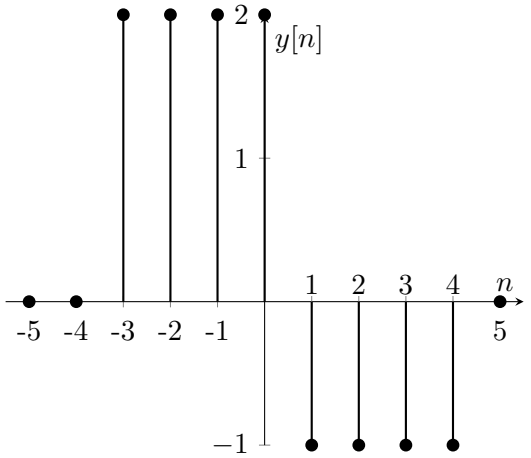
   (a) Sketch the signal $y[n]$ where $y[n] = x[-n+3]$.

   (b) Sketch the signal $z[n]$ where $z[n] = x[2n+1]$.

3. For each of the following plots of a discrete-time signal, express the signal in terms of **(i)** the Kronecker delta $\delta[n]$ and **(ii)** the unit-step signal $u[n]$. All samples not shown may be assumed to be zero.




(a)




(b)

4. For each of the following parts, **(i)** determine the value(s) of $z$ that satisfy the equation and **(ii)** plot these values of $z$ on the complex plane. **Hint:** recall that complex numbers in polar form may be written as $Re^{j\theta} = Re^{j(\theta_p + 2\pi k)}$ where $\theta_p$ is the *principal angle* and $k$ is some integer.

   (a) $z^4 - 1 = 0$

   (b) $z^4 + 1 = 0$

5. A clipping system is a common block included in larger digital systems. For constants $a$ and $b$ where $a < b$, the clipping system with input $x[n]$ produces output $y[n]$ as follows.

$$y[n] = \begin{cases} b, & x[n] \geq b \\ x[n], & a < x[n] < b \\ a, & x[n] \leq a \end{cases}$$

   (a) Let $x[n] = \{-4, -3, -2, -1, \underset{\uparrow}{0}, 1, 2, 3, 4\}$, $a = -2$, and $b = 2$. Sketch $y[n]$.

   (b) Determine if the clipping system is linear or non-linear. Please provide justification, e.g. proof, counter-example, explanation.

   (c) Determine if the clipping system is time-invariant or time-varying. Please provide justification, e.g. proof, counter-example, explanation.

6. Another common operation in digital systems is known as *windowing* or applying a *window function*. We will discuss window functions in much more detail later in the course. For now, we can think of window functions as extracting $N$ samples in a given range of indices and discarding samples outside this range. For an input $x[n]$ and output $y[n]$, a windowing system is described as follows.

$$y[n] = \begin{cases} x[n], & 0 \leq n \leq N-1 \\ 0, & \text{otherwise} \end{cases}$$

   (a) Let $x[n] = \{-2, -1, \underset{\uparrow}{0}, 1, 2, 3, 4, 5, 6\}$. Sketch $y[n]$ for $N = 4$

   (b) Determine if the windowing system is linear or non-linear. Please provide justification, e.g. proof, counter-example, explanation.

   (c) Determine if the windowing system is time-invariant or time-varying. Please provide justification, e.g. proof, counter-example, explanation.